MATH 2101 Linear Algebra I, FALL 2025

Tutorial 7 Solution

1. (a) Note that

$$[T(u_1 - u_2 + u_3)]_{\beta'} = [T]_\beta^{\beta'}[u_1 - u_2 + u_3]_\beta = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ -1 & 1 & -2 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix}.$$

So $T(u_1 - u_2 + u_3) = 3u_1 - (u_1 + u_2) - 4(u_1 + u_2 + u_3) = -2u_1 - 5u_2 - 4u_3$.

(b) Let $[\mathrm{Id}_U]_\beta^{\beta'} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$. Then

$$u_1 = au_1 + d(u_1 + u_2) + g(u_1 + u_2 + u_3)$$
$$u_2 = bu_1 + e(u_1 + u_2) + h(u_1 + u_2 + u_3)$$
$$u_3 = cu_1 + f(u_1 + u_2) + i(u_1 + u_2 + u_3),$$

equivalently,

$$(a + d + g - 1)u_1 + (d + g)u_2 + gu_3 = 0$$
$$(b + e + h)u_1 + (e + h - 1)u_2 + hu_3 = 0$$
$$(c + f + i)u_1 + (f + i)u_2 + (i - 1)u_3 = 0.$$

As u_1, u_2, u_3 are linearly independent,

$$a + d + g - 1 = d + g = g = 0$$
$$b + e + h = e + h - 1 = h = 0$$
$$c + f + i = f + i = i - 1 = 0,$$

and hence $[\mathrm{Id}_U]_\beta^{\beta'} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

(c) Let $\gamma = \{w_1, w_2, w_3\}$. Then

$$u_1 = w_1 - w_2$$
$$u_2 = w_2 - w_3$$
$$u_3 = w_2,$$

and hence

$$w_1 = u_1 + w_2 = u_1 + u_3$$
$$w_2 = u_3$$
$$w_3 = w_2 - u_2 = u_3 - u_2.$$

Thus, $\gamma = \{u_1 + u_3, u_3, u_3 - u_2\}$.

2. (a) Proof: If $[x]_\beta = [y]_\beta = (a_1, a_2, \cdots, a_n)^T$, then

$$x = a_1u_1 + \cdots + a_nu_n = y.$$

(b) Proof: Let $\gamma = \{w_1, w_2, \cdots, w_n\}$. For any $v \in V, [v]_\beta = [v]_\gamma$. Then for any $j = 1, 2, \cdots, n$,

$$\begin{aligned} [w_j]_\beta &= [w_j]_\gamma \quad \text{by taking } v = w_j \\ &= \mathbf{e}_j \\ &= [u_j]_\beta, \end{aligned}$$

and hence by (a), $w_j = u_j$. Thus, $\beta = \gamma$.

3. (a) Consider the equation

$$c_1Sv_1 + c_2Sv_2 + \ldots + c_kSv_k = 0$$

for some scalars c_j $(j = 1, \cdots, k)$, it follows from the invertibility of S that

$$c_1v_1 + c_2v_2 + \ldots + c_kv_k = 0.$$

As $v_1, \cdots, v_k$ are linear independent, we have $c_1 = c_2 = \cdots = c_k = 0$.

(b) For any $x \in \mathbb{R}^n$,
($\supset$): if $x \in \text{span}(\gamma)$, then $x = a_1Sv_1 + \cdots + a_kSv_k$ for some $a_1, \cdots, a_k \in \mathbb{R}$. Note that

$$a_1v_1 + \cdots + a_kv_k \in \text{span}(\beta) = N(B).$$

Thus

$$Ax = AS(a_1v_1 + \cdots + a_kv_k) = SB(a_1v_1 + \cdots + a_kv_k) = 0 \text{ and hence } x \in N(A).$$

($\subset$): if $x \in N(A)$, then

$$\begin{aligned} Ax = 0 &\implies BS^{-1}x = S^{-1}Ax = 0 \implies S^{-1}x \in N(B) = \text{span}(\{v_1, \cdots, v_k\}) \\ &\implies x = \text{span}(\{Sv_1, \cdots, Sv_k\}) = \text{span}(\gamma). \end{aligned}$$

4. (a) No. By definition, the zero vector cannot be an eigenvector.

(b) Yes. Suppose $Au = \lambda u$ for some $u \in \mathbb{R}^3 \setminus \{0\}$ and $\lambda \in \mathbb{R}$. Then

$$A(2u) = 2(Au) = 2(\lambda u) = \lambda(2u).$$

As $2u \neq 0$, it is an eigenvector of A.

(c) No. For example, consider $A = 0_{3\times3}$, $u = (3, 0, 0)^T$ and $v = (-2, 0, 0)^T$, then u and v are eigenvectors of A (corresponding to the eigenvalue $\lambda = 0$), but $3u + 2v = 0$ is not.